



# International Technology Roadmap for Wide Bandgap Power Semiconductors: A Five Year Review

## Introduction

**T**he first edition of the International Technology Roadmap for Wide Bandgap Power Semiconductors (ITRW) was published in September 2019 [1]. This was a watershed moment for the IEEE Power Electronics Society (PELS) being the first time that a global roadmap had been attempted by PELS, especially across such a diversity of both technology and applications. Five years on is an ideal time to look back with some perspective on the projections, predictions and reality of wide bandgap (WBG) power semiconductors, its adoptions across a variety of sectors, and in particular look at the questions raised in the first ITRW edition and how they may have changed. The global context since ITRW was published in 2019 also highlights how global events can have a dramatic impact on technological developments with a global pandemic starting just a few months after the publication of ITRW, and its consequences remaining in place for several years. This impacted not only technological work at the research and development stage,

but also severely degraded supply chains leading to serious issues of device and module delivery. World events including conflicts and the increasing impact of climate change also had a significant impact on key drivers for electrification, with a consequential push for even more integration of WBG bandgap power devices, more rapidly. It is not overstating the case to say that WBG based power electronics are now well in the mainstream across many technology areas and applications, with many practical challenges being solved or mitigated by engineers across the globe. This article will review some of the key questions raised in ITRW, provide a perspective on how well we did with ITRW in predicting the penetration of WBG power semiconductors and identify lessons that can be learned for such roadmapping efforts in the future.

## People and Roadmaps

Before looking to the detailed technical questions raised by ITRW in 2019, it is worth considering the impact on the people who participated in the roadmap exercise and the wider power electronics community. When the ITRW effort started in late 2015, and seriously in 2016, the group was

very small (less than 10 people worldwide). However, this rapidly grew into a global community of power electronics researchers and engineers looking to understand how this new technology could be realized, what research challenges needed to be addressed, and how the devices could be deployed in practical applications. The use of open workshops with a very inclusive approach, i.e., everyone was welcome from students and young professionals, through to leading researchers and industrialists, regardless of location and background. This led to a diversity of thought, ideas and perspectives that was essential to capture more than just the voice of a small group of specialists. This community-based approach also had a hugely positive impact with workshops across the globe, and bringing in a much wider network of collaborating organizations and people. Some of the latter were working on their own technology roadmaps, such as on packaging and materials, for example, but brought great practical ideas, supportive efforts and a bilateral communication into related fields of interest that enhanced the ITRW work, and broadened perspectives beyond power electronics. The

community effort also had a strong aspect of community building within the power electronics society. This approach brought new individuals into the society, fostered greater engagement and activities that went beyond the roadmap itself and filtered into technical committees, building bridges between diverse technical communities (such as between device and application specialists, who often were not aware of the competing requirements and constraints in their respective fields.

This wider community persists, with personal and professional contacts made that are stronger than would have been without ITRW, and lasting links were forged with benefits extending far beyond the publication of the roadmap. The overarching purposes of the IEEE

and PELS to “Advance Technology for Humanity,” and “Powering a sustainable future for the benefit of humanity,” respectively, are captured in the ITRW team ethos. In writing this section about people, it is impossible not to recognise and celebrate the life and contributions of Braham (Bram) Ferreira whose vision of a WBG technology roadmap came to fruition in his lifetime. Without his dedication, drive and vision to initiate ITRW, when he was president of PELS, and to trust others to deliver the roadmap, ITRW would not have happened.

It is fair to say that the process of developing roadmaps is not always easy. The International Technology Roadmap for Semiconductors (ITRS) was one of the key roadmaps in shaping the future vision of the

semiconductor industry. It was initially very focussed on small geometries and the drive to maintain “Moore’s Law” in digital integrated circuits (ICs). Latterly, ITRS broadened its scope to include a variety of technologies that Silicon (and other materials) offered, but as the physical limits of Silicon were approaching, it became less and less of a focus for the industry. Nevertheless, the importance of maintaining a clarity of vision meant that ITRS needed a new home, and the IEEE stepped up to host the ITRS activities. As ITRW was developing its own community and effort going forward, the IEEE was also finding its own path to roadmap success. The ITRW found itself as the trail-blazer with an active community and clarity of purpose that became

**Table 1. ITRW 2019 Predictions and Status—Silicon Carbide and Gallium Nitride devices.**

| Working Group | Prediction                                                                                                                                                                                                                                                                                                      | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SiC           | 200 mm SiC wafer material will be commercially available. Defects and wafer planarity will still need improvement. Foundries will process 200 mm SiC wafers in their Si lines for economy of scale.                                                                                                             | <b>Achieved</b><br>200 mm wafers are close to commercial availability from multiple vendors. Today, 200 mm wafers are processed in a dedicated SiC only fab. Several Si fabs are about to start fabrication using 200 mm SiC wafers. Defects and planarity are still an issue. A competitive market is driving costs down.                                                                                                                                   |
| SiC           | The price of SiC MOSFET devices will be within 2 times that of its similarly rated Si counterparts. SiC devices will be inserted in 10% EV/HEV, and PV as the two top applications. 3.3 kV SiC MOSFET devices will become commercially available. 6.5 kV SiC MOSFET devices will be close to commercialization. | <b>Achieved</b><br>The price of SiC MOSFETs is within 2-3 times that of its similarly rated Si counterparts for 1200 V MOSFETs with lower current ratings. EV/HEV, and PV are the two top applications. EV/HEV insertion is estimated to have met the 10% target. 3.3 kV SiC MOSFET became commercially available in 2022 and are now offered by several vendors. 6.5 kV MOSFETs engineering samples available but no commercial release from major vendors. |
| GaN           | GaN power HEMTs in commercial production on 8-inch Si wafers processed on CMOS-compatible lines.                                                                                                                                                                                                                | <b>Achieved</b><br>GaN power devices are now being commercially produced on 8-inch Si wafers.                                                                                                                                                                                                                                                                                                                                                                |
| GaN           | Normally-off operation achieved by techniques such as p-type gates at the device level or cascode configuration at the package level.                                                                                                                                                                           | <b>Achieved</b><br>Normally-off GaN HEMTs available in cascode configuration and p-type gate from multiple vendors, respectively.                                                                                                                                                                                                                                                                                                                            |
| GaN           | Voltage ratings up to 900 V available and in competition with Si and SiC devices at comparable cost.                                                                                                                                                                                                            | <b>Achieved</b><br>900 V GaN power HEMTs are commercially available.                                                                                                                                                                                                                                                                                                                                                                                         |
| GaN           | Research on devices with higher voltage ratings in progress, likely up to 1200 V, in competition with SiC.                                                                                                                                                                                                      | <b>Achieved</b><br>In practice, SiC still dominates the 1200 V voltage space and GaN has 1250 V products from at least one supplier. Rigorous research on devices with 1200 voltage rating in progress and engineering samples available from several vendors.                                                                                                                                                                                               |
| GaN           | Research ongoing on package-level gate drive integration (heterogeneous), as well as chip-level fully integrated gate drives (homogeneous small-scale ICs).                                                                                                                                                     | <b>Achieved</b><br>The chip-direct approach and focus on package integration of GaN devices has largely been realized.                                                                                                                                                                                                                                                                                                                                       |

a role model for the IEEE roadmap efforts going forward, in partnership with others. Since ITRW was published, PELS has pushed forward with new roadmaps such as ITRG (International Technology Roadmap for High-Power Electronics for Modern Energy Grids), and ITRD (International Technology Roadmap of Power Electronics for Distributed Energy Resources). The IEEE now has a technology roadmaps committee (IRC) that brings roadmap leaders together to share experiences and good practice, building a wider global community (<https://roadmaps.ieee.org/>). This is one of the strongest legacy aspects of ITRW that has a long-lasting impact.

### Reflections on the ITRW 2019 Five Year Market Predictions

The market projection for silicon carbide (SiC) in 2023, at the time of writing the ITRW roadmap, was U.S. \$1.5 billion. Recent data shows that the SiC market was significantly higher than the predictions, with some market analysts showing the global SiC market being U.S. \$3.72 billion in 2023 (Source: Precedence Research, 2024) and projected to be more than U.S. \$4 Billion in 2024. Figures for gallium nitride (GaN) have been slightly lower than the U.S. \$423 million projected for 2023, with figures varying from U.S. \$260 million to U.S. \$326 million (Sources: Yole, 2024 and Straits Research, 2024). However the growth of GaN technology is projected in some analyses to be at a rate of 35% to 2030 (Source: Global Market Insights, 2024) leading to a dramatic growth in this sector. These market figures show that ITRW was too conservative on SiC, and over-optimistic for GaN. The growth of electrification of transport has been a huge driver for SiC technology and this explains its rapid growth. The under-estimation of the growth of GaN technology is perhaps due to the impact of the pandemic given its stage of development at that time.

Nevertheless, the growth of GaN technology seems to be recovering rapidly now as various technical challenges have been solved.

The broad predictions of ITRW 2019 of significant growth across WBG power semiconductors turned out to indeed be the case, with perhaps SiC outperforming the market predictions the most. The broad questions that the ITRW roadmap was designed to focus on were defined as follows:

- 1) What are the decisive benefits for the adoption of WBG?
- 2) Can the investment costs be justified by the long-term benefits?
- 3) Which applications will benefit first from the adoption of WBG?
- 4) What is the threshold to move to WBG?

In addition, the individual ITRW working groups were asked to report back to the ITRW committee with the following specific technical questions:

- 1) How should your area advance in the next five-, 10- and 20-year time-frames?
- 2) What major roadblocks do you foresee? Any technological issues?
- 3) Name one killer application for your area? (Opinion)
- 4) Any specs that you see changing or are needed to advance your area?
- 5) How disruptive will these new technologies be? What impact will they have? What is the broader impact?
- 6) How do you see the cost structure changing?

These questions provided a framework for consistency across technology and application areas and scope for both quantitative and qualitative perspectives.

Given the five-year time frame since ITRW was launched in 2019, it is interesting to focus on the five year advance predictions and make a judgement about the status of each in 2024 [1], as depicted in Tables 1 and 2.

### Summary

The predictions of the five-year horizon from ITRW 2019 have shown to

be generally accurate, and in particular the identified “killer application” of SiC devices in transport electrification and GaN devices in charging and consumer electronics have become a reality. The market predictions have been surpassed in the case of WBG power semiconductors, with SiC in particular making extensive inroads into many application areas for voltages above 650 V. The ITRW roadmap still provides significant value in guiding ongoing and future research, and the questions used in ITRW 2019 are still valid when looking ahead to the next five-, 10- and 20-year horizons. Finally, in thinking about lessons from the road-mapping process, the biggest takeaway from the ITRW experience is less about the accuracy of the predictions (although so far they have turned out to be pretty good!), and more about bringing the power electronics community together to work towards a common goal across different technology and geographical areas. This has resulted in us achieving a unity of purpose in building a community and sharing results across our society for the benefit of our society membership and the wider power electronics field as a whole.

### About the Author

**Peter Wilson**, Ph.D., M.B.A., B.Eng. (Senior Member, IEEE) ([p.r.wilson@bath.ac.uk](mailto:p.r.wilson@bath.ac.uk)) is currently a Full Professor of electronic engineering with the University of Bath, U.K. He is also serving as an IEEE PELS VP for Industry and Standards and Member-at-Large. He has held academic positions as the Deputy Director of the Dyson Institute of Engineering and Technology, Dyson, Malmesbury, U.K., a Professor and the Head of Electronic and Electrical Engineering at the University of Bath, a Lecturer and a Reader at the University of Southampton, U.K., and with industry roles at Analogix Inc. and Ferranti plc. His research interests include modelling and simulation, magnetics, design automation (spe-

**Table 2. ITRW 2019 Predictions and Status—Packaging, Integration and Applications.**

| Working Group                           | Prediction                                                                                                                                                                                                                                                                                                                                                                                                                                    | Status                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SiC System Level Applications           | SiC will have large market shares in automotive, aviation, railway, industry drives, renewable energy and utility-based power electronics applications such as solid-state transformers.                                                                                                                                                                                                                                                      | <b>Partially Achieved</b><br>SiC has become the power electronics technology of choice for automotive, railway, industry drives, and low voltage (3.3 kV and below) renewable energy applications. Aviation electrification penetration is slow due to the stringent qualification standards. Utility-based power electronics insertion is slow due to limited high-voltage device (6.5 kV, and 10 kV) availability. |
| System Integration                      | The power system will at least find SiC application in such demonstrations projects in less than five years as power electronics transformers, solid static substations or flexible substations.                                                                                                                                                                                                                                              | <b>Partially Achieved</b><br>SiC has become the power electronics technology of choice for all these areas. Achieved for applications requiring 3.3 kV and lower voltage devices. At higher voltage levels, the limited high-voltage device (6.5 kV, and 10 kV) availability is delaying commercial system insertion.                                                                                                |
| System Integration and Application: GaN | In the next five years, GaN will have large market penetration in LIDAR, datacenter, consumer power electronics (charger and adapter), and drones.                                                                                                                                                                                                                                                                                            | <b>Largely Achieved</b><br>GaN has become widespread in 650 V and below power converter applications, and has penetrated the datacenter, consumer power electronics and chargers markets. Silicon is still widely used in drones due to its low cost, however GaN is making inroads into this area.                                                                                                                  |
| WBG Packaging and Integration           | Pressure from users for familiar packages in a like-for-like replacement format will continue to drive the bulk of packaging for modules (with consequent limitations in performance) and for discrete dies. Innovation in packages for discrete dies and modules, mainly for niche applications, will see more widespread introduction of direct bonded Cu foil as an interconnect, replacing wire bonds and providing lead-out connections. | <b>Achieved</b><br>It has been noticeable how many WBG devices have adopted standard packages such as TO220 or TO247 and modules with standard pin arrangements used in many applications. Wire bonds are being displaced or reengineered to achieve the low parasitic inductance that enables efficient high frequency operation.                                                                                   |
| WBG Packaging and Integration           | Integration of additional functionality will be limited to small-value ceramic decoupling capacitors, sensors and potentially gate drives (subject to temperature constraints).                                                                                                                                                                                                                                                               | <b>Achieved</b><br>This has been specifically the case for gate drive circuits and ICs, sensors and small value ceramic capacitors. In particular, GaN devices are co-packaged with intelligent gate drives and ICs with prognostic, diagnostic, and safety features.                                                                                                                                                |
| WBG Packaging and Integration           | Chip-direct packages will be a focus for experienced users of GaN devices at lower power levels and for higher frequency switching. It is unlikely that there will be widespread use of chip-direct packages for SiC.                                                                                                                                                                                                                         | <b>Achieved</b><br>The use of direct-chip packages has become standard practice for GaN. However, SiC uses packaging techniques similar to those of conventional Silicon devices.                                                                                                                                                                                                                                    |

cifically computational intelligence techniques including genetic algorithms and machine learning), systems engineering, robotics, open-source tools and methods, and embedded systems. He was the Chair of the ITRW Roadmap, IEEE Std 1573, and has published widely in

power electronics and related fields with more than 150 articles, several patents, authored a number of books, and received the IEEE Standards Medallion in 2020. He is a Fellow of IET, the British Computer Society, and the Higher Education Academy.

## Reference

- [1] *International Technology Roadmap for Wide Bandgap Power Semiconductors*, IEEE Power Electron. Soc., 2019, doi: 10.17023/cgfc-4692. [Online]. Available: <https://resourcecenter.ieee.org/roadmap/pelspro0020>

